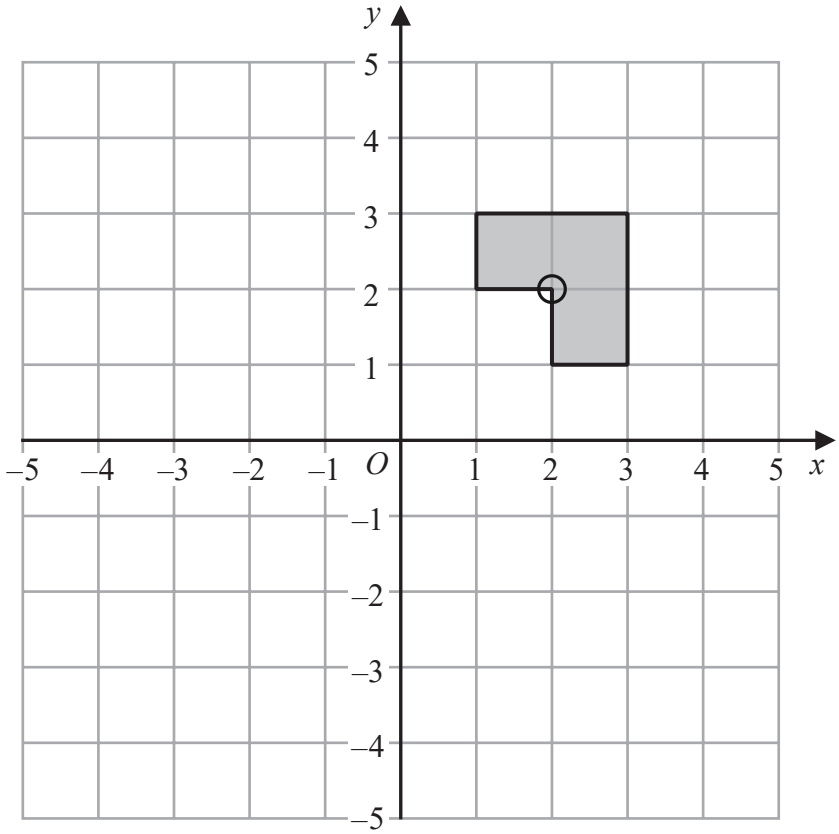
21



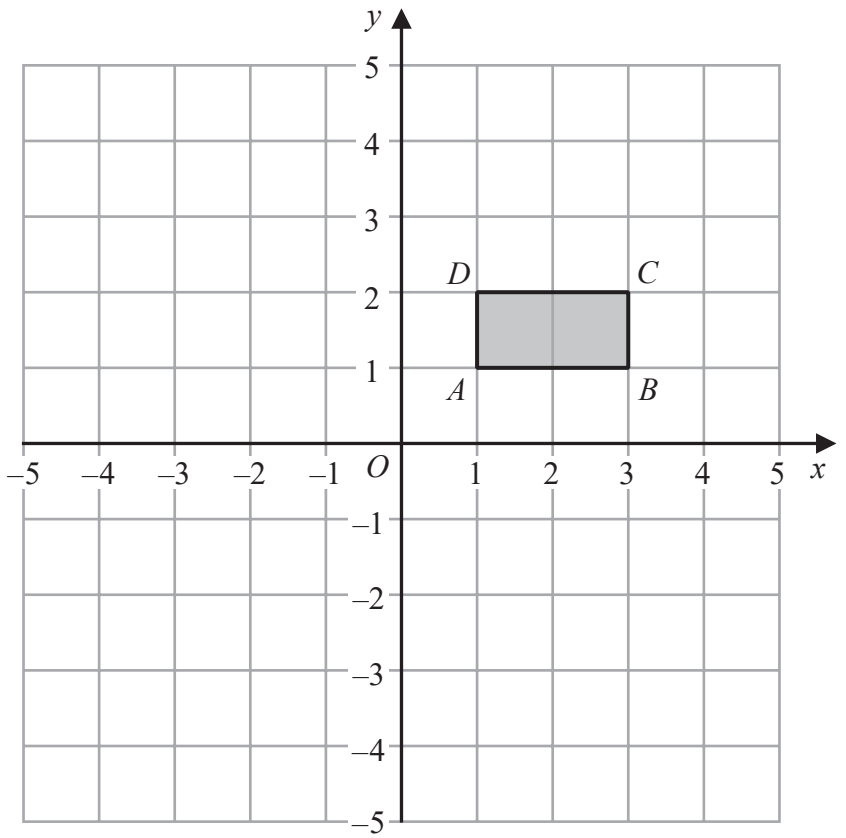
The shaded shape is rotated 180° about the point (2, 2)

(a) How many of the vertices of the shaded shape are invariant?

1 vertex: (2, 2), the centre of rotation.

1 mark
Only the centre stays fixed.

(1)



Rectangle $ABCD$ is transformed by a combination of two transformations so that

all points on AB are invariant
and there are no other invariant points.

The first transformation is

a reflection in the line with equation $y = k$, where k is an integer and $k \neq 1$

(b) Describe fully the second transformation.

A point (x,1) first maps to (x,2k – 1).

Second transformation: translation by $\begin{bmatrix} 0 \\ 2-2k \end{bmatrix}$

1 mark
Translation.

1 mark
Column vector (0, 2−2k).

(2)

(Total for Question 21 is 3 marks)